Practice Exam - 1

Question 10 of 117

A patient with a history of hypertension, hyperlipidemia, and HFrEF (LVEF: 25%, NYHA III) presents to the emergency department complaining of a one-day history of excruciating chest pain, sudden in onset and described as sharp and stabbing. The initial workup revealed normal renal and liver function, normal cardiac troponin, and an elevation in inflammatory markers. His medications consist of the following sacubitril/valsartan, dapagliflozin, digoxin, rosuvastatin, aspirin, amlodipine, metoprolol succinate. He is diagnosed with acute pericarditis and colchicine is started.

Which of the following medications carries the highest risk of causing a serious drug-drug interaction with colchicine?

A. Digoxin

B. Metoprolol

C. Rosuvastatin

D. Amlodipine

Commentary | References

Testing point: The Board-certified AHFTC specialist should identify potential drug-drug interactions and adjust medical therapy as needed in patients with HFrEF.

Correct Answer: A. Digoxin

Rationale: Colchicine primarily undergoes liver metabolism via cytochrome P450 3A4 (CYP3A4) followed by hepatobiliary elimination. In addition, about 10-20% of the drug is renally eliminated using the permeability glycoprotein ATP efflux transporter (P-gp). In addition to being responsible for the renal excretion, the P-gp system regulates the concentration of colchicine in plasma by modulating the gastrointestinal absorption of the drug. Many cardiovascular drugs interact with the CYP3A4 and/or the P-gp efflux system therefore careful revision of medications should occur before prescribing colchicine. This is important as drug-drug interactions can lead to either early colchicine discontinuation or toxicity.

Digoxin (Answer A) is a substrate of both intestinal and renal P-gp. Further, rhabdomyolysis has been reported with concomitant use of digoxin and colchicine, as digoxin can interfere with the clearance of colchicine leading to muscle or bone marrow toxicity, thus making Answer A the most correct.

Metoprolol succinate (Answer B) is primarily metabolized via CYP2D6 and does not have a clinically significant interaction with colchicine. Although statins, including rosuvastatin (Answer C), may increase the risk of myopathy or rhabdomyolysis when combined with colchicine, and this **potential interaction is noted in the FDA prescribing information for rosuvastatin**, the risk is generally lower and less clinically significant than the P-gp–mediated interaction seen with digoxin. Rosuvastatin FDA Prescribing Information (https://www.accessdata.fda.gov/drugsatfda_docs/label/2023/021366s043s044lbl.pdf?utm_source=chatgpt.com)

Dihydropyridine calcium channel blockers, such as amlodipine (Answer D), do not inhibit CYP450 enzymes or P-gp and can be safely co-administered with colchicine, making Answer D incorrect.